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Software-Defined Networking (SDN)

Short definition: SDN is a networking approach where control is done by software in a central controller instead of being embedded separately inside every switch/router.

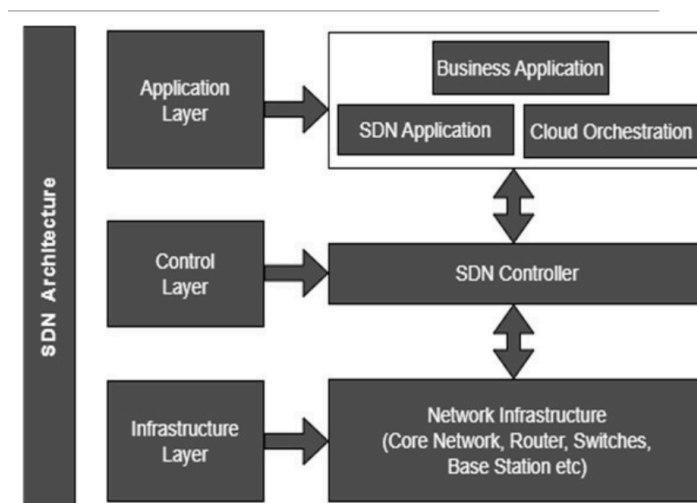
SDN stands for **Software-Defined Networking**.

The main idea is to separate the **control plane** from the **data plane**.

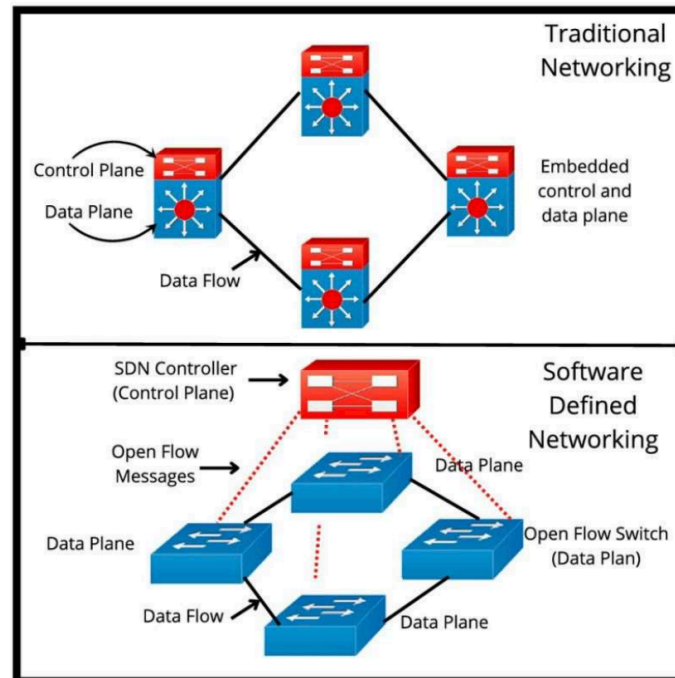
- **control plane** = decides where traffic should go
- **data plane** = actually forwards packets

In traditional networking, both are usually built into each router or switch.

In SDN, the control logic is moved to a **central controller**, while the switches mainly forward traffic according to rules sent by that controller.



SDN vs Traditional Networking



Traditional Networking

In traditional networks:

- each router or switch has its own control plane and data plane
- each device makes forwarding decisions locally
- configuration is often done box-by-box

So the network is more distributed, but also harder to manage as it grows.

SDN

In SDN:

- the **SDN controller** acts as the control plane
- switches mainly act as forwarding devices in the data plane
- the controller communicates with switches using control messages such as **OpenFlow** messages

instead of every switch providing its own logic, the controller has a wider view of the whole network.

What this means physically:

- traditional networking = each device decides for itself
- SDN = one logical controller tells devices how to behave

Key Advantages

Centralized Control

SDN controllers manage the entire network from a single point.

This improves:

- efficiency
- automation
- easier network-wide decision making

Programmability

Network behavior can be modified via APIs.

So the network becomes easier to adapt when requirements change.

Improved Traffic Management

Dynamic routing can optimize traffic based on real-time conditions.

This helps the network react better to congestion or changing traffic loads.

Scalability

New devices can be added more easily without complicated manual configuration on every box.

Cost Reduction

SDN can reduce operational costs by automating tasks that would otherwise require manual configuration.